

Measuring Total Factor Productivity in China with Data Factor Inputs: A Non-Parametric Machine Learning-Based Stochastic Frontier Analysis

摘 要

关键词

Abstract

Keywords

1 Introduction

Total Factor Productivity (TFP) has long been an important indicator for measuring the quality of economic growth. However, since the 1980s, the economics community has extensively discussed the "Solow Paradox." Solow (1987) pointed out that despite the rapid development of information technology, represented by computers, TFP growth has not increased significantly. This phenomenon contradicts the traditional understanding of the relationship between technological progress and economic growth. In the traditional economic growth accounting framework, TFP measurement mainly relies on production factors such as capital and labor, neglecting the role of data as a factor of production. This omission may be one of the key reasons behind the Solow Paradox, as the results obtained fail to accurately reflect the contribution of data factors. In recent years, with the rapid development of the digital economy, data has gradually emerged as a crucial factor in economic growth, contributing to increased production efficiency, optimized resource allocation, and reduced information asymmetry (Jones and Tonetti, 2020).

In 2022, the "Opinions of the State Council of the Central Committee of the Communist Party of China on Establishing a Data Infrastructure System to Better Leverage the Role of Data Factors" (hereinafter referred to as the "Opinions") officially recognized data as the fifth production factor alongside land, labor, capital, and technology. It also called for "accelerating the development of the data factor market." Data factors have thus become a key driver of high-quality economic development. Therefore, reassessing TFP by incorporating data factors into the economic growth analysis framework is crucial for accurately measuring their actual contribution.

This paper makes two key contributions: First, it constructs a TFP measurement framework incorporating data factors based on the non-parametric machine learning method of Stochastic Frontier Analysis (SFA) (Wei, 2024). Compared with traditional TFP measurement methods, the non-parametric machine learning SFA model captures systematic and nonlinear relationships between production factors and output using Bayesian Additive Regression Trees (BART). This overcomes the strong assumptions of traditional parametric approaches regarding production function forms and improves the accuracy of TFP measurement by learning the production frontier in a data-driven manner. Second, by integrating data factors into the production framework, this study re-examines their actual contribution to TFP. This not only provides a new explanatory perspective on the Solow Paradox but also offers empirical support for the marketization of data factors, thereby promoting the further development of the data factor market.

The structure of this paper is as follows: Section 2 reviews relevant literature, summarizing research on the impact of data factors on TFP; Section 3 introduces the research methodology and data sources, explaining in detail the advantages of using the Bayesian Additive Regression Tree Stochastic Frontier Model to measure TFP; Section 4 presents the empirical analysis, estimating TFP with data factors and exploring its contribution to economic growth; Section 5 summarizes the research conclusions and provides policy recommendations.

2 Literature Review

2.1 Research on Factor Inputs in TFP

2.2 Methods for Measuring TFP

3 Research Methodology and Data Sources

3.1 Bayesian Additive Regression Tree Stochastic Frontier Model

The Bayesian Additive Regression Tree Stochastic Frontier Model is a method that combines Bayesian parameter estimation with Stochastic Frontier Analysis (SFA) for estimating the production frontier. The Bayesian approach integrates prior knowledge and provides uncertainty measures for inefficiency estimation. Bayesian Additive Regression Trees (BART) are a class of machine learning decision tree methods used to model complex, non-parametric relationships between response variables and a set of predictor variables (Chipman et al., 2010). The main idea is to approximate complex high-dimensional functions using an additive structure of multiple regression trees, generating the production frontier through self-learning.

The functional representation of the production frontier model based on Bayesian Additive Regression Trees is as follows:

$$Y_i = \beta_0 + \sum_{j=1}^m g_j(\mathbf{x}_i; T_j, M_j) + V_i - U_i \quad (1)$$

where Y_i represents the output of province i , $g_j(x_i; T_j, M_j)$ is the predicted value from the j -th decision tree, representing a BART decision tree model. The BART decision tree model consists of the combination of input variables $\mathbf{x}_i$, the decision tree structure T_j , and the decision tree leaf node parameters M_j , where $M_j = \{\mu_{1,j} \dots \mu_{b,j}\}$ and b is the number of leaf nodes in the j -th tree. The term $V_i \sim N(0, \sigma_v^2)$ represents measurement errors, and $U_i \sim HN(0, \sigma_u^2)$ follows a half-normal distribution, indicating technical inefficiency.

BART decision tree models operate in two main stages: 1) Establishing the prior probability. 2) Using posterior probability to obtain each tree's predictions and summing them to derive the final prediction results.

3.1.1 Determination of Prior Probability

In BART, the prior probability plays a key role in controlling model complexity, preventing overfitting, and ensuring stable sampling. The prior distribution is defined as:

$$p((T_1, M_1), \dots, (T_m, M_m), \sigma_u, \sigma_v, \beta_0) = \left[\prod_{j=1}^m p(M_j | T_j) p(T_j) \right] p(\sigma_v) p(\sigma_u) p(\beta_0) \quad (2)$$

where all tree structures T_j and leaf node values M_j are assumed to be independent, as well as independent from $\sigma_u, \sigma_v, \beta_0$. The prior distribution of tree structure $p(T_j)$ controls the complexity of decision trees in BART-SFM, preventing overfitting. The prior of leaf node weights $p(M_j | T_j)$ ensures reasonable predictions for each tree. The prior distributions $p(\sigma_v)$ and $p(\sigma_u)$ regulate the variability of errors, while the prior $p(\beta_0)$ models the overall data level, ensuring both flexibility and stability of the BART-SFM model.

3.2 Posterior Inference

To estimate the production frontier, BART-SFM employs MCMC sampling, using a combination of Gibbs sampling and the Metropolis-Hastings algorithm to estimate posterior distributions. The target posterior conditional distribution is:

$$p((T_1, M_1), \dots, (T_m, M_m), \beta_0, \sigma_v^2, \sigma_u^2, U | \text{data}). \quad (3)$$

Each sampled (T_j^*, M_j^*) generates a production frontier estimate $f^*(\cdot)$, and the final production frontier estimate is obtained by averaging over these samples.

The estimation process of the BART decision tree model is illustrated in Figure 1.

3.3 Total Factor Productivity Estimation

According to the decomposition method of Total Factor Productivity (TFP) proposed by Ru Shaofeng and Wei Boyang (2018) and Kumbhakar (2000), TFP can be decomposed

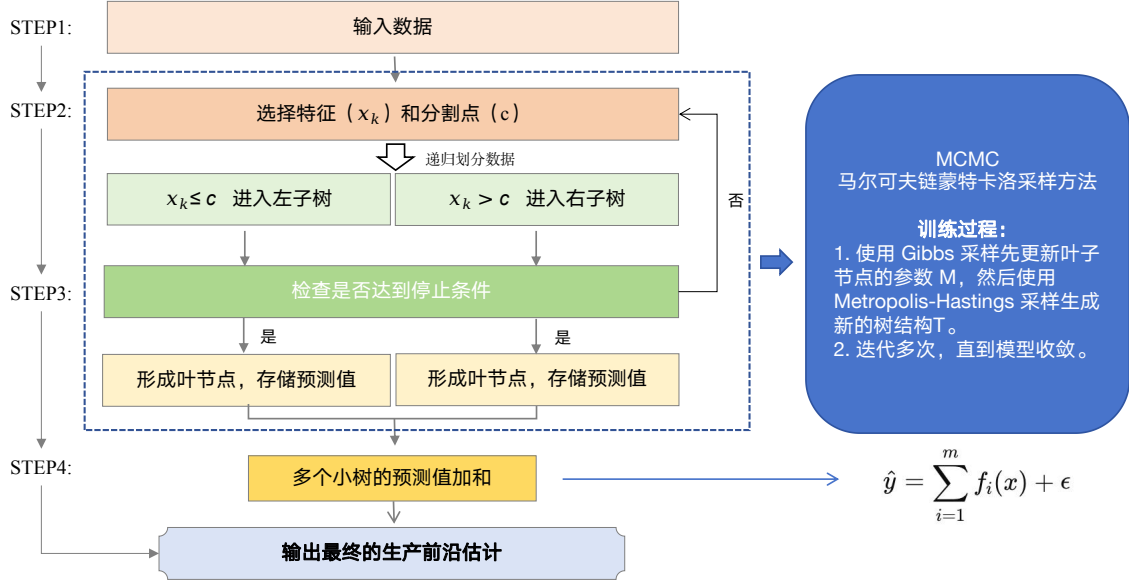


图 1 Flowchart of the Bayesian Additive Regression Tree Stochastic Frontier Model

into the rate of technical change (TC), the rate of technical efficiency change (TEC), and the rate of scale efficiency change (SEC). The underlying principle is to subtract the growth rates of various inputs (typically weighted by cost shares) from the total output growth rate, thereby attributing TFP growth to technological progress, scale effects, and changes in technical efficiency.

Let $\Delta \ln X_i$ denote the growth rate of input i , then the output growth can be expressed as:

$$\Delta \ln Y_{it} = \frac{\partial \hat{f}^{(b)}(\mathbf{x}_{it})}{\partial t} \Delta t + \sum_{j=1}^N \theta_{j,it} \Delta \ln \mathbf{x}_{j,it} + (\Delta v_{it} - \Delta u_{it}) \quad (4)$$

Here, $\theta_{j,it} = \frac{\partial \hat{f}^{(b)}(\mathbf{x}_{it})}{\partial \ln \mathbf{x}_{j,it}}$ is the output elasticity with respect to input j .

The growth rate of Total Factor Productivity (TFP) is defined as the output growth rate minus the input growth rates weighted by their respective cost shares s_i , i.e.,

$$\Delta \ln TFP_{it} = \Delta \ln Y_{it} - \sum_{j=1}^N s_{j,it} \Delta \ln \mathbf{x}_{j,it} \quad (5)$$

where $s_{j,it}$ is the cost share of input j in total cost or total expenditure (satisfying $\sum_{j=1}^N s_{j,it} = 1$). Random noise (Δv_{it}) is generally assumed to have zero expectation in the mean. After rearranging, we obtain the decomposition formula of TFP change:

$$\Delta \ln TFP = \underbrace{\frac{\partial \hat{f}^{(b)}(\mathbf{x}_i)}{\partial t} \Delta t}_{\text{Technical Change (TC)}} + \underbrace{\left[\left(\sum_{j=1}^N \theta_{j,it} - 1 \right) \Delta \ln X_{it} \right]}_{\text{Scale Efficiency Change (SEC)}} - \underbrace{\Delta u}_{\text{Technical Efficiency Change (TEC)}} \quad (6)$$

Here, X_{it} denotes the composite growth rate of inputs. In practical computation, due to differences in growth rates among factor inputs, input growth is weighted by the relative elasticities w_j , where $\Delta \ln X_{it} = \sum_{j=1}^N w_j \Delta \ln \mathbf{x}_{it}$, and $w_j = \theta_j / \sum_{j=1}^N \theta_j$. Regarding scale efficiency, when the sum of output elasticities $\sum_{j=1}^N \theta_{j,it}$ is not equal to 1, it implies non-constant returns to scale. In this case, input expansion (or contraction) results in additional output changes, which is referred to as the scale effect. Δu represents the change in the inefficiency term u (i.e., technical efficiency). When technical efficiency improves, u decreases, making $-\Delta u$ a positive value. Here, $\Delta u = TE_{it} - TE_{it-1}$.

The technical efficiency based on BARTSFA can be written as:

$$TE_{it} = \frac{1}{B} \sum_{b=1}^B E \left(e^{-U} \mid \hat{f}^{(b)}(\mathbf{x}_{it}), \hat{\sigma}_v^{2,(b)}, \hat{\sigma}_u^{2,(b)} \right) \quad (7)$$

where B is the number of samples. The Rao-Blackwell estimator is used to compute the posterior mean of TE. This ultimately yields the technical efficiency for each decision-making unit and evaluates its performance relative to the production frontier. The value of TE ranges from 0 to 1, with higher values indicating that the firm is closer to the production frontier.

4 TFP Measurement Results for China

5 Contribution of TFP to Economic Growth in China

6 Conclusions and Policy Recommendations